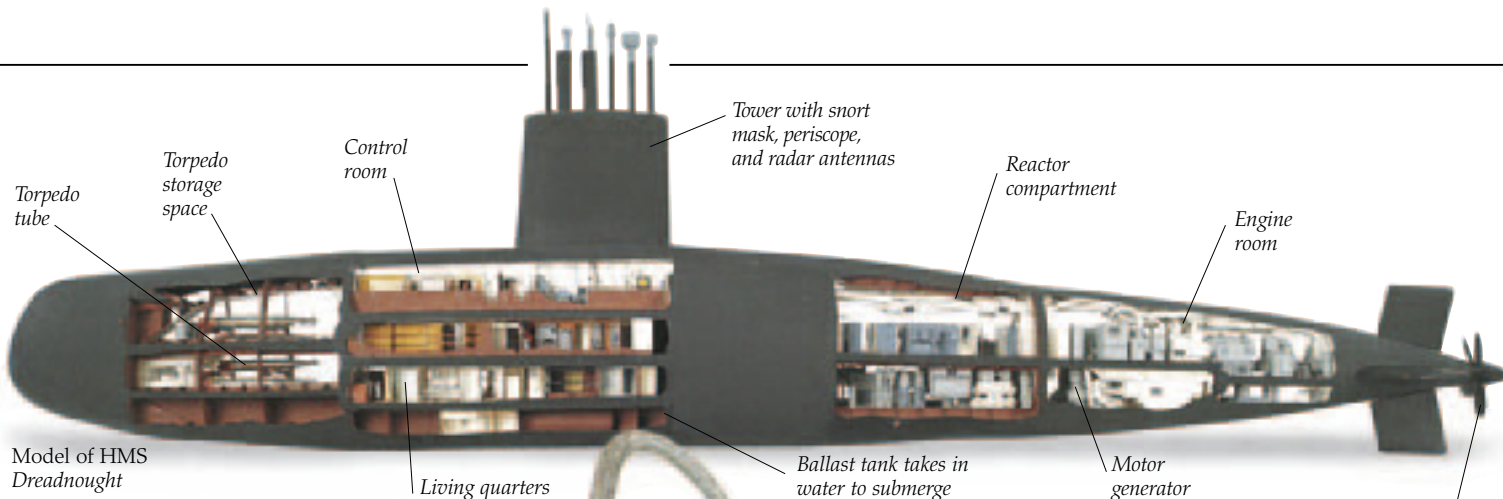
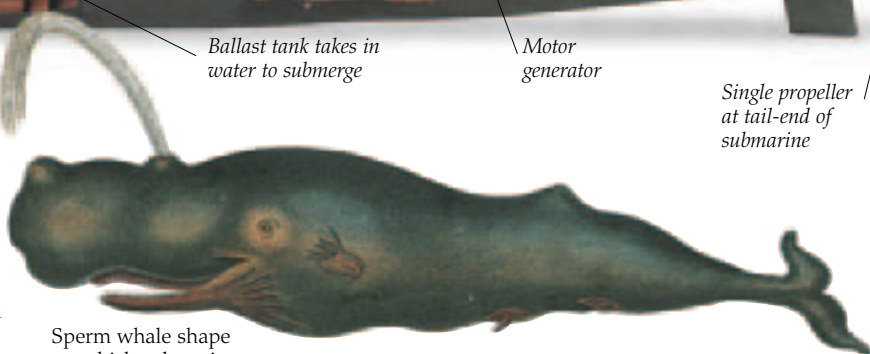




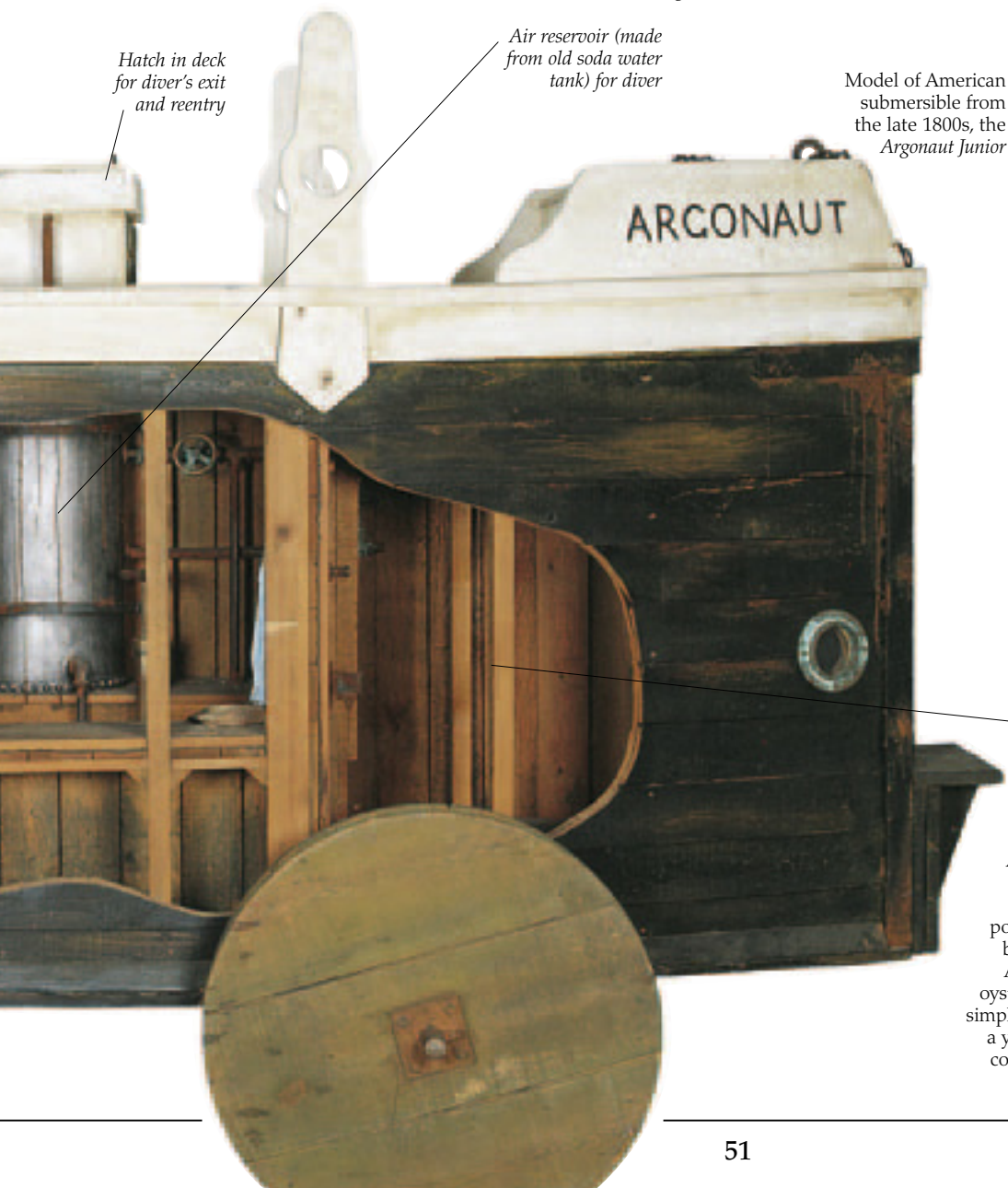
Model of HMS Dreadnought

WHALE-SHAPED

HMS *Dreadnought* has only a single propeller at its tail-end and is shaped like a whale. It also has a snort mast, in the same way that the whale has a blow-hole on the top of its head. However, a whale must surface to breathe air, while HMS *Dreadnought* has an air-conditioning system so its 88-person crew can spend weeks submerged. Launched in 1960, it was Britain's first nuclear-powered submarine and the first submarine to surface at the North Pole. She was decommissioned in 1981 after cracks appeared in her hull and cooling systems.



Sperm whale shape on which submarine design is based



Model of American submersible from the late 1800s, the Argonaut Junior



INSIDE STORY

The interior of a submersible shows how cramped it can be with only just enough room for a pilot and perhaps one or two passengers. On deep dives, it can get quite cold inside.

Compartment with air lock to maintain enough pressure for diver

BOTTOM CRAWLER

This model of an underwater vehicle, called the *Argonaut Junior*, was built in New York in 1894. It was designed by Simon Lake who later went on to design other submarine craft. *Argonaut Junior*, powered by a hand-crank, was driven over the sea bed. It only had a maximum depth of 20 ft (6 m). A helmeted diver could leave the craft to pick up oysters, clams, and other objects. The helmet was of simple construction made from canvas and steel with a yacht's porthole used for a window. A metal tank contained the diver's air supply and a garden hose supplied air from the tank to the helmet.